TELECOM CUSTOMER CHURN PREDICTION PROJECT

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INTRODUCTION

- TelecomCorp is experiencing significant customer attrition, losing valuable revenue. The need for a predictive model is urgent to identify at-risk customers before they leave.

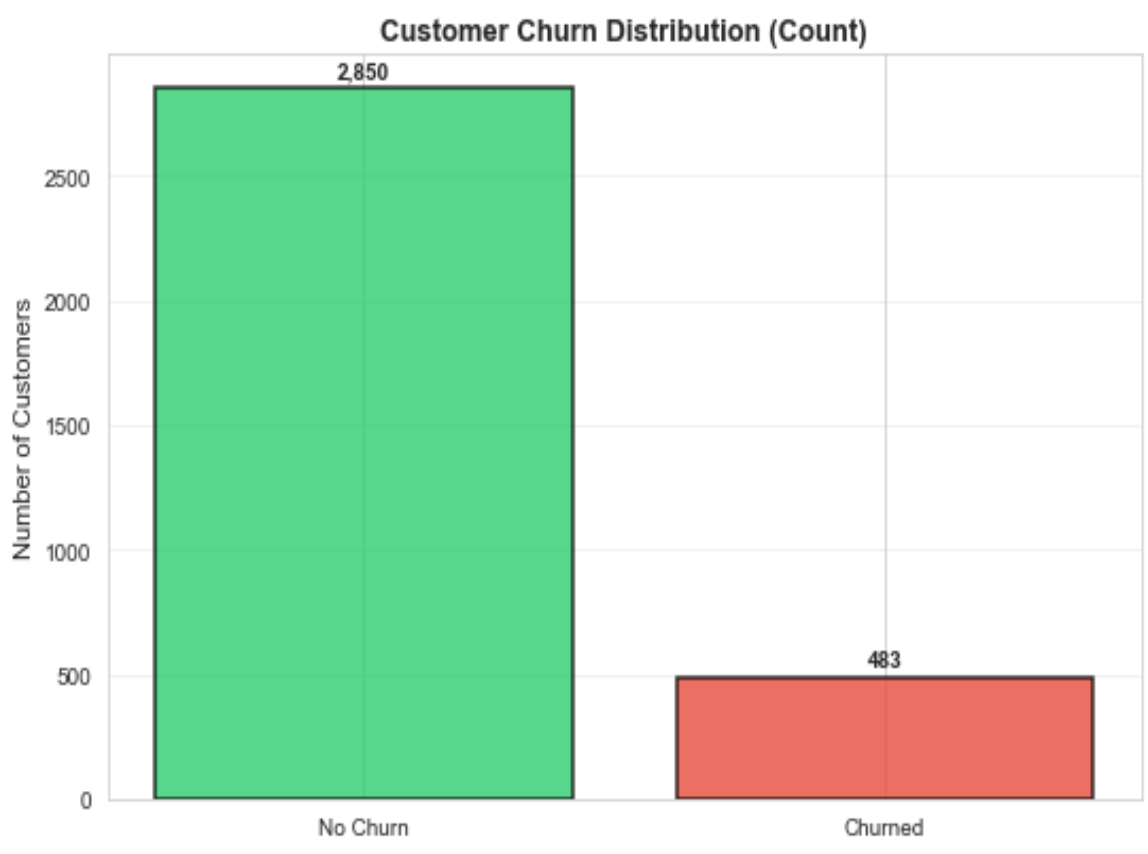
Business Problem

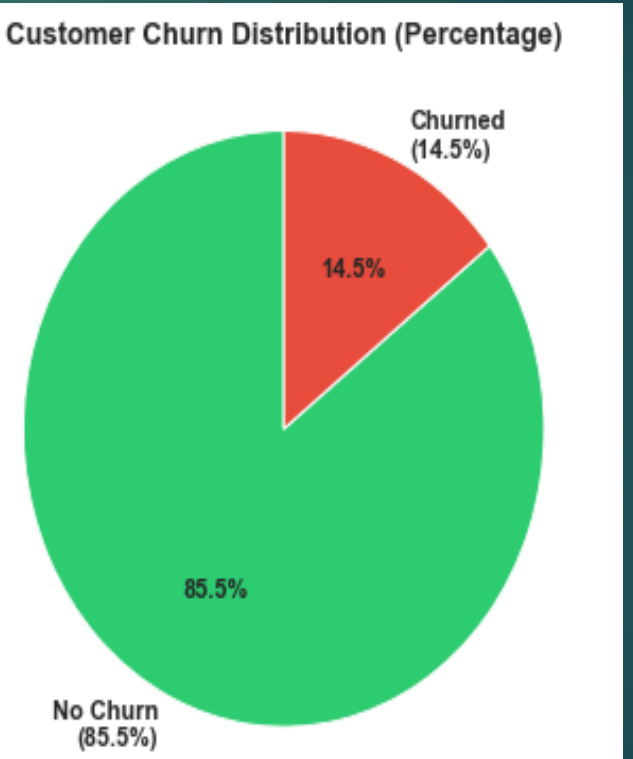
- TelecomCorp is losing customers due to a 14.5% annual churn rate, and they have no systematic way to identify which customers are about to leave before it happens.

Objectives

- Build a baseline logistic regression model
- Optimize the model through hyperparameter tuning
- Compare using classification metrics (Precision, Recall, F1)
- Identify model providing best business value
- Provide implementation recommendations

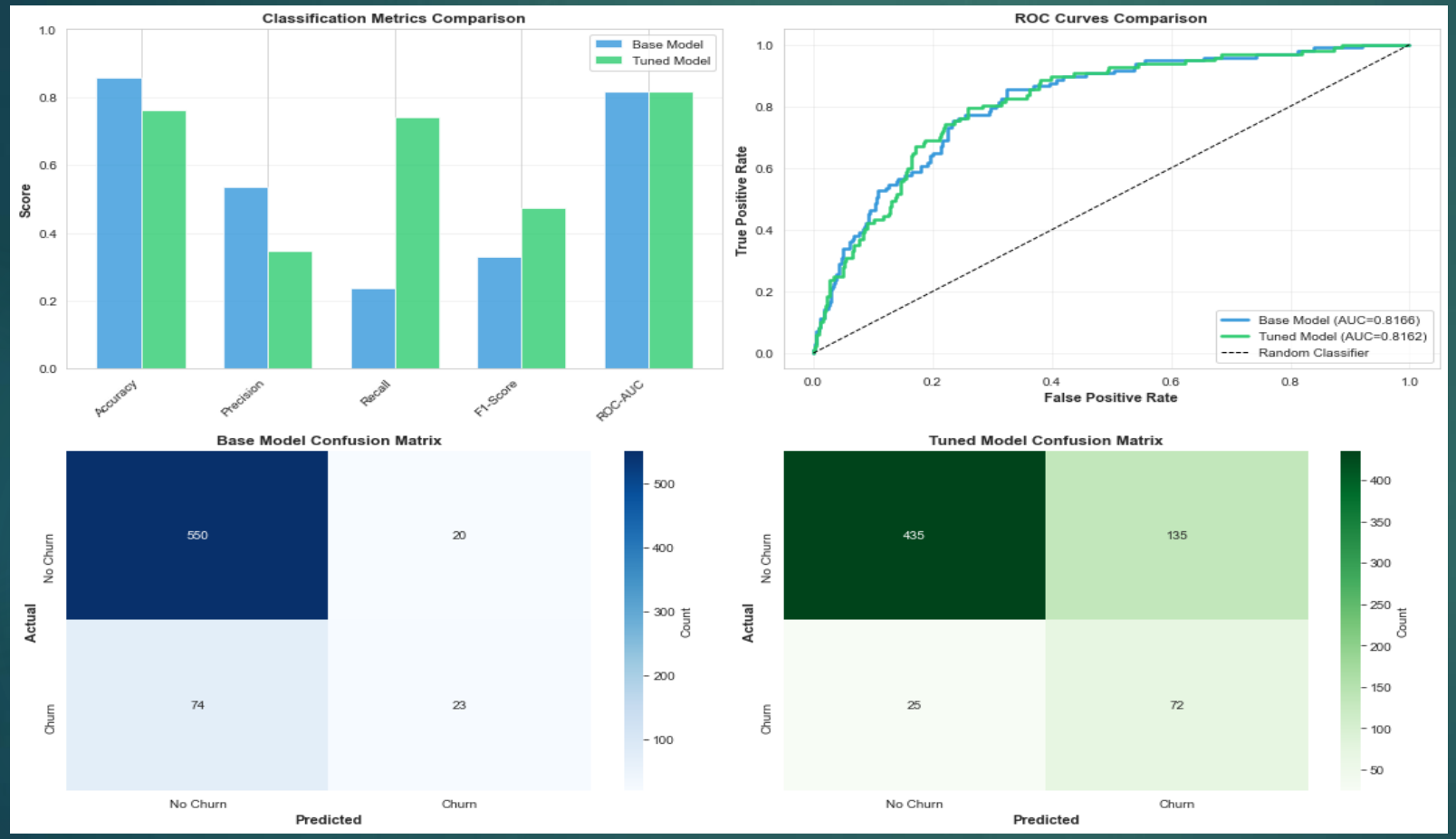
Findings through visuals





Here we get to see the Imbalance between churn and No churn.

Second Visual: Model Comparison



In this visual we get to see the tuned model outperforms the base model across most evaluation metrics (Accuracy, Precision, Recall, F1-Score, and ROC-AUC).
Improvements indicate better overall predictive performance and more balanced classification.

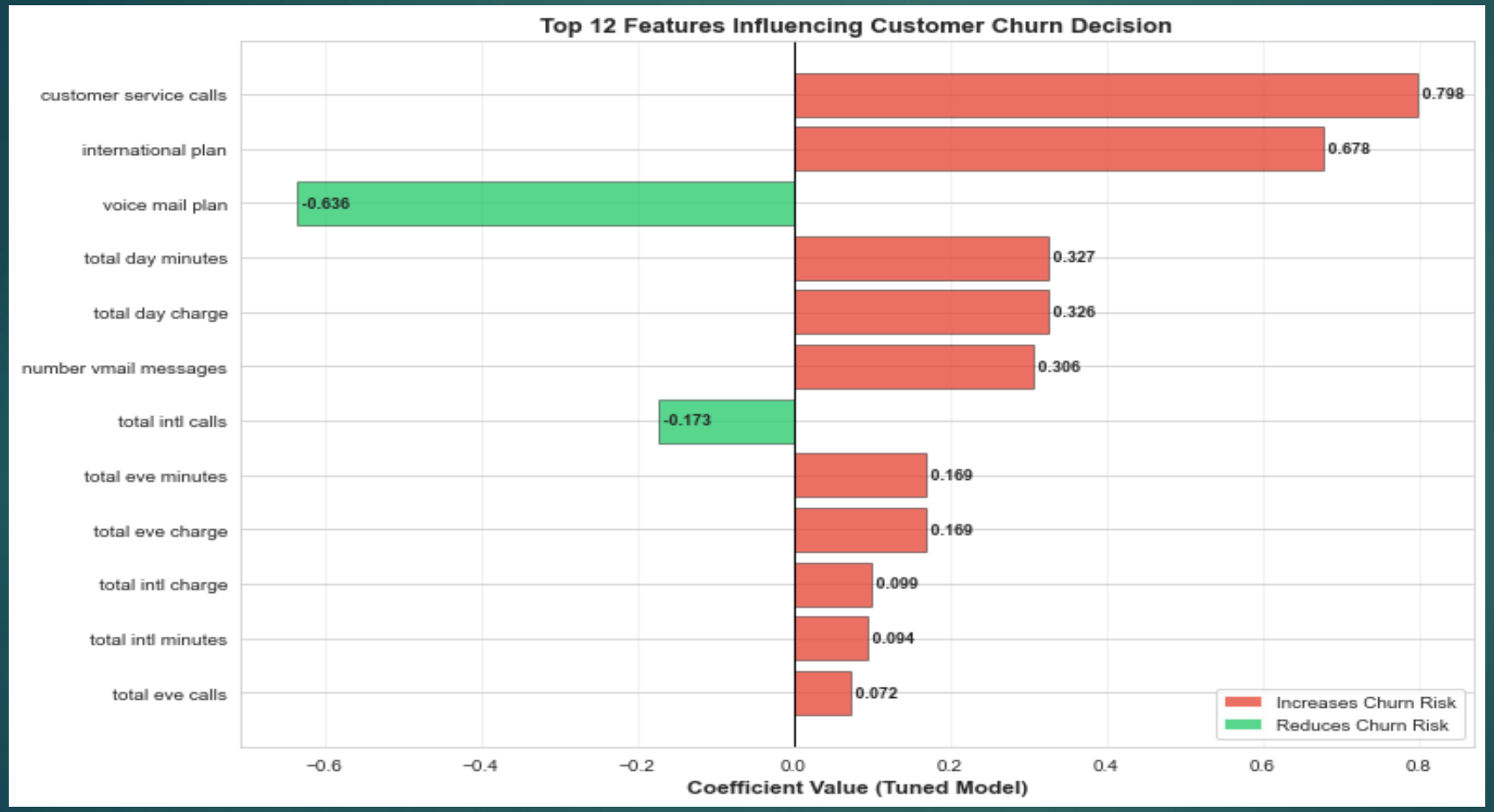
ROC Curve Insights

- The tuned model's ROC curve is closer to the top-left corner, showing stronger discrimination ability.
- Higher AUC score confirms the tuned model is better at distinguishing between churn and non-churn customers.
- Both models perform better than a random classifier, but the tuned model is clearly superior.

Confusion Matrix Analysis

- Base Model:
  - Higher number of misclassifications (false positives and false negatives).
  - Less reliable in identifying churn cases.
- Tuned Model:
  - Reduced misclassification errors.
  - Improved detection of actual churn cases (better recall).
  - More accurate overall predictions.

Last Visual: Feature Importance



The chart highlights the top 12 features influencing churn decisions in the tuned model.
Red bars (positive coefficients) - Increase likelihood of churn
Green bars (negative coefficients) - Reduce likelihood of churn

Explanation:

- **Top Increasing Churn Risk**
- Frequent customer service calls - Strong indicator of dissatisfaction
- Lack of international plan - Customers more likely to leave
- High daytime usage - Possible service quality or pricing concerns
- **Factors Improving Customer Retention**
- Voice mail plan subscription - Strongly linked to customer loyalty
- Higher international call activity - Engaged, retained customers

Recommendations

- Deploy Tuned Model to score all customers monthly
- Create Risk Segments based on churn probability
- Launch Retention Campaigns targeting top 100-150 at-risk customers

Conclusion

- PRODUCTION-READY:
- The tuned model successfully handles the 85:15 class imbalance and delivers actionable insights. By focusing on recall (catching churners) over accuracy, I've built a system aligned with business objectives and ready to drive measurable retention improvements.